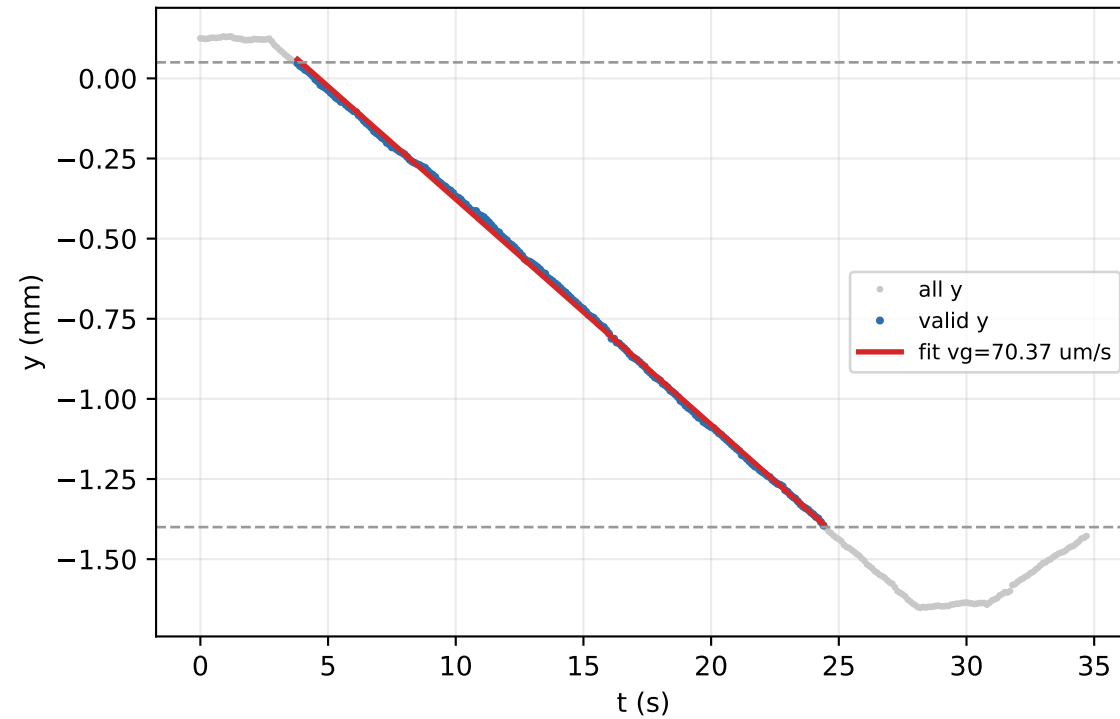
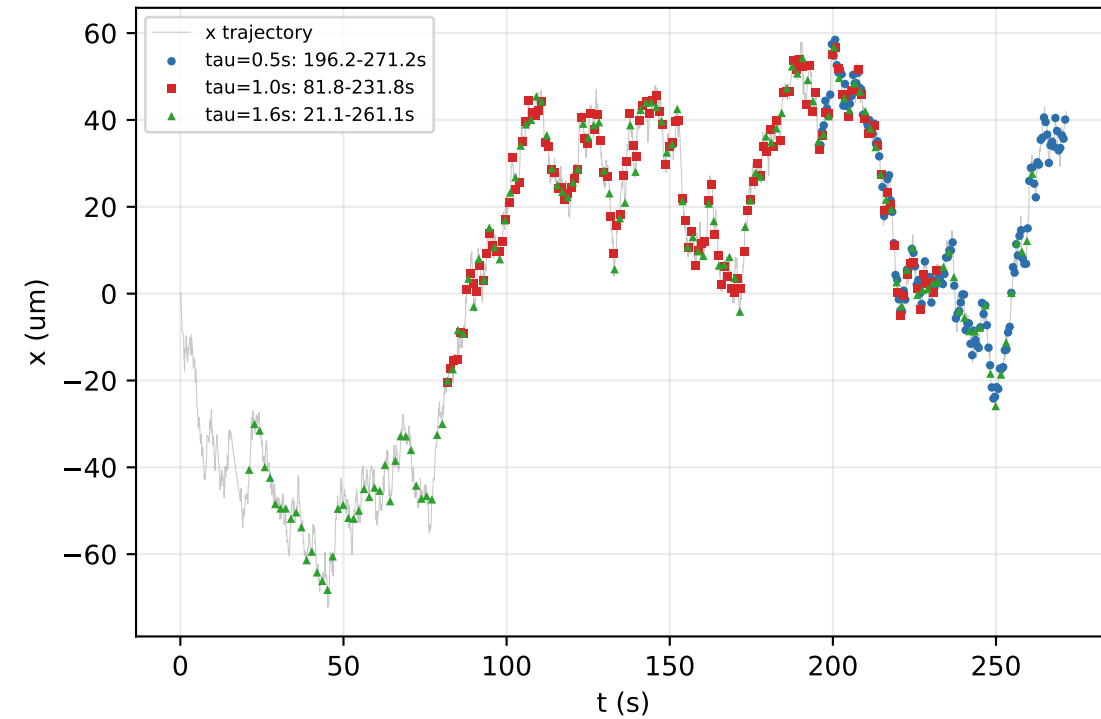


Millikan oil drop particle3: optimized Brownian intervals

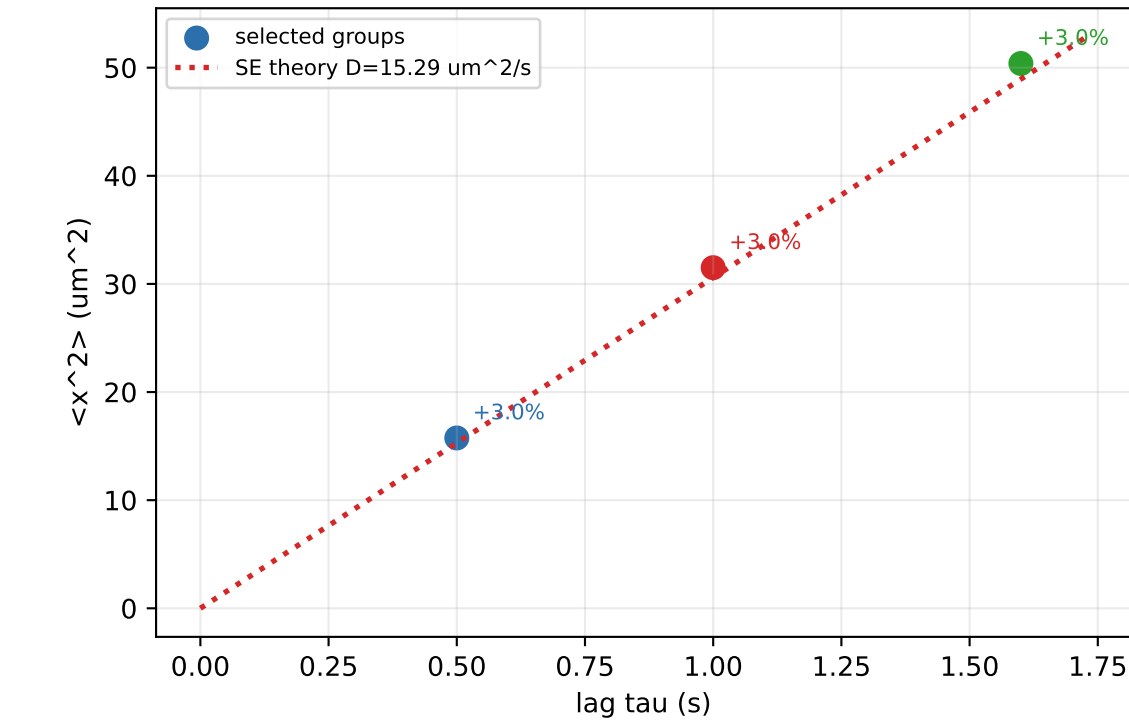
Free-fall terminal velocity



Sampled coordinate points per lag group



Separate $\langle x^2 \rangle$ results



Summary

Particle: particle3
 Window strategy: target_kb
 Freefall valid points: 207
 $v_g = 70.3666 \text{ um/s}$, $R^2 = 0.999526$
 $r_{\text{eff}} = 0.7781 \text{ um}$, $d_{\text{eff}} = 1.5561 \text{ um}$
 SE theory $D = 15.2938 \text{ um}^2/\text{s}$

tau	covered time	$\langle x \rangle$	$\langle x^2 \rangle$	k_B err
0.5s	196.2-271.2s	0.039	15.753	+3.00%
1.0s	81.8-231.8s	0.172	31.504	+3.00%
1.6s	21.1-261.1s	0.454	50.388	+2.96%

NIST $k_B = 1.3806\text{e-}23 \text{ J/K}$
 No average is reported.